

Exam #2

Math 742 - Mathematical Statistics

3/25/2026

Name David Cortes

Directions: The in-class portion of exam 1 is worth 50 points. The exam is split into 2 sections. Part I (problems 1-3) is 50 minutes. Part II (problems 4-5) is 25 minutes. You may have a hand held calculator for the arithmetic and a 8 1/2 x 11" sheet (one-sided) of notes.

Part I - Theory + asymptotics

Problem 1 (10 points)

Let $X_1, X_2, \dots, X_n$ be an iid sample from an normal distribution with mean μ and variance $\sigma^2 < \infty$. Consider the estimator,

$$\bar{T} = 2\bar{X}$$

- a) (3 points) Find the bias of T as an estimate of μ .
- b) (4 points) Find $\text{Var}(T)$.
- c) (3 points) Compute the mean squared error of T and determine whether T is consistent for μ

We are given $\bar{T} = 2\bar{X}$ where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Since $X_1, \dots, X_n$ are iid with $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$

We know $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

a) Start with expectation: $E[T] = E[2\bar{X}]$

Use linearity of expectation: $E[T] = 2E[\bar{X}]$

since $E[\bar{X}] = \mu \rightarrow E[T] = 2\mu$

Now Apply the bias definition:

$$\text{Bias}(T) = E[T] - \mu$$

Substitute $E[T] = 2\mu$

$$\text{Bias}(T) = 2\mu - \mu = \mu$$

So $T = 2\bar{X}$ is biased for μ , except in the special case $\mu = 0$

Page intentionally left blank

Part (b) find $\text{Var}(T)$

Start with $\text{Var}(T) = \text{Var}(2\bar{X})$

Use variance scaling: $\text{Var}(2\bar{X}) = 2^2 \text{Var}(\bar{X}) = 4 \text{Var}(\bar{X})$

Since $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, we get $\boxed{\text{Var}(T) = 4 \cdot \frac{\sigma^2}{n}}$

Part (c)

Use the identity $\text{MSE}(T) = \text{Var}(T) + (\text{Bias}(T))^2$

We got already $\text{Var}(T) = \frac{4\sigma^2}{n}$, $\text{Bias}(T) = \mu$

So $\text{MSE}(T) = \frac{4\sigma^2}{n} + \mu^2 \rightarrow \text{Thus } \boxed{\text{MSE}(T) = \frac{4\sigma^2}{n} + \mu^2}$

Is T consistent for μ ?

T_n is consistent for μ if $T_n \xrightarrow{P} \mu$

Here, $T = 2\bar{X}$. Since the sample mean is consistent $\bar{X} \xrightarrow{P} \mu$, multiplying $\times 2$ gives $2\bar{X} \xrightarrow{P} 2\mu$. But our estimator is supposed to estimate μ , not 2μ . Therefore $T \not\xrightarrow{P} \mu$
so T is not consistent for μ , except in the special case $\mu=0$, because then $2\mu=\mu=0$

We can also see this from the $\text{MSE}(T) = \frac{4\sigma^2}{n} + \mu^2$. As $n \rightarrow \infty$, $\frac{4\sigma^2}{n} \rightarrow 0$, so

$\text{MSE}(T) \rightarrow \mu^2$. This does not go to 0 unless $\mu=0$. That also shows the estimator is not consistent in general.

2. Let $X_1, X_2, \dots, X_n$ be an iid sample from $N(\mu, \sigma^2)$, where $\sigma^2 < \infty$ is known.

a) (3 points) Find the Fisher information $I_n(\mu)$ for this sample.

b) (3 points) State the Cramér-Rao lower bound for unbiased estimators of μ .

c) (4 points) Is $\bar{X}$ efficient? Justify your answer using variance comparisons and interpret what efficiency means in this setting.

(a) Fisher Information $I_n(\mu)$.

The likelihood is $L(\mu) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$

So the log-likelihood is $l(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$

Differentiate: $\frac{\partial l(\mu)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$ and $\frac{\partial^2}{\partial \mu^2} l(\mu) = -\frac{n}{\sigma^2}$

Therefore, $I_n(\mu) = -E\left[\frac{\partial^2 l(\mu)}{\partial \mu^2}\right] = \frac{n}{\sigma^2} \rightarrow \boxed{I_n(\mu) = \frac{n}{\sigma^2}}$

(b) CRLB

For any unbiased estimator T of μ , $\text{Var}(T) \geq \frac{1}{I_n(\mu)}$

Since $I_n(\mu) = \frac{n}{\sigma^2}$, $\text{Var}(T) \geq \frac{\sigma^2}{n}$

So the CRLB is:

$$\boxed{\text{Var}(T) \geq \frac{\sigma^2}{n}}$$

(c) Since $E[\bar{X}] = \mu$

The sample mean is unbiased for μ . Also, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$.

This equals the Cramér-Rao lower bound. Therefore, $\bar{X}$ attains the minimum possible variance among all unbiased estimators of μ , so $\boxed{\bar{X} \text{ is efficient}}$

3. Let $X_1, X_2, \dots, X_n$ be an iid sample with mean μ and variance $\sigma^2 < \infty$.

Define:

$$T_n = \frac{\sqrt{n}(\bar{X} - \mu)}{S_n}$$

Where S_n^2 is the sample variance.

- a) (3 points) State the Central Limit Theorem by specifying the limiting distribution of the properly standardized sample mean $\bar{X}$.
- b) (3 points) Explain what it means for an estimator to be consistent. Then explain why S_n^2 is a consistent estimator of σ^2 .
- c) (4 points) Use Slutsky's Theorem to determine the limiting distribution of T_n .

(a) CLT

Since $X_1, \dots, X_n$ are iid with mean μ and variance $\sigma^2 < \infty$, the CLT states

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \quad \text{equivalently,} \quad \sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

(b) Consistency and consistency of S_n^2

An estimator W_n of a parameter θ is consistent if $W_n \xrightarrow{P} \theta$.

This means the estimator converges in probability to the true parameter value as $n \rightarrow \infty$.

The sample variance is a consistent estimator of the population variance,

$$\text{so } S_n^2 \xrightarrow{P} \sigma^2$$

(c) Limiting distribution of T_n

from part (a),

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

since $S_n^2 \xrightarrow{P} \sigma^2$, the continuous Mapping Theorem implies

$$S_n = \sqrt{S_n^2} \xrightarrow{P} \sigma$$

Therefore,

Page intentionally left blank

$$\frac{S_n}{\sigma} \xrightarrow{P} 1$$

Now rewrite T_n as

$$T_n = \frac{\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}}{\frac{S_n}{\sigma}}$$

The numerator converges in distribution to $N(0, 1)$, and the denominator converges in probability to 1.

Then by Slutsky's theorem:

$$\boxed{T_n \xrightarrow{d} N(0, 1)}$$

Part II — Hypothesis Testing

Problem 4

Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2 = 4)$

We want to test:

$$H_0: \mu = 0 \text{ vs } H_1: \mu > 0$$

a) (3 points) Construct an appropriate standardized test statistic.

Because the X_i 's are iid Normal, then $\rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

Since $\sigma^2 = 4$,

$$\bar{X} \sim N\left(\mu, \frac{4}{n}\right)$$

under the null hypothesis $H_0: \mu = 0$, this becomes

$$\bar{X} \sim N\left(0, \frac{4}{n}\right)$$

Now standardize $\bar{X}$.

The mean under H_0 is 0, and the standard deviation under H_0 is

$$\sqrt{\frac{4}{n}} = \frac{2}{\sqrt{n}}$$

So the standardized test statistic is

$$Z = \frac{\bar{X} - 0}{\frac{2}{\sqrt{n}}}$$

Therefore under H_0 ,

$$Z \sim N(0, 1)$$

So the answer to part (a) is

$$Z = \frac{\bar{X}}{2/\sqrt{n}} = \frac{\sqrt{n} \bar{X}}{2}, \quad \boxed{}$$

$$Z \sim N(0, 1)$$

Part (b) Give the rejection region for a level $\alpha = 0.05$ test.

Because the alternative is

$$H_1: \mu > 0,$$

Large positive values of $\bar{X}$ support H_1 . Therefore we reject H_0 for large values of Z .

So the rejection has the form

$$\text{Reject } H_0 \text{ if } Z > c$$

for some cutoff c .

Now we determine c from the level condition.

$$\text{A level } \alpha = 0.05 \text{ test means } P(\text{reject } H_0 \mid H_0 \text{ true}) = 0.05$$

Since under H_0 , $Z \sim N(0, 1)$, this becomes $P(Z > c) = 0.05$

Now rewrite it in terms of the standard normal CDF Φ :

$$P(Z > c) = 0.05 \iff P(Z \leq c) = 0.95 \iff \Phi(c) = 0.95$$

Therefore,

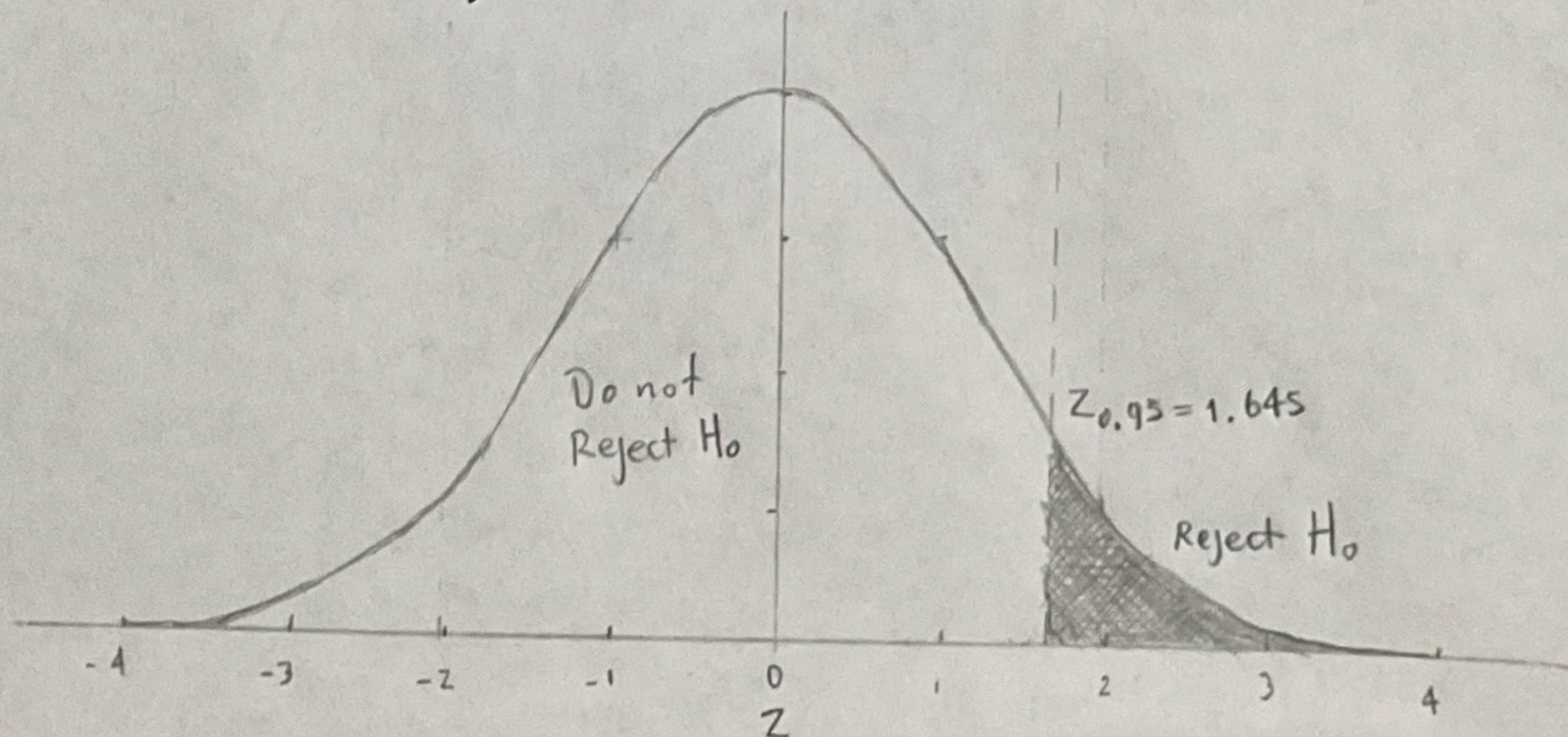
$$c = \Phi^{-1}(0.95)$$

Using the standard normal table or calculator, (From 18_Hypothesis Testing_Details.pdf)
Rejection Region (page 2)

$$c = \Phi^{-1}(0.95) = 1.645$$

Thus the rejection region is

$$\text{Reject } H_0 \text{ if } Z > 1.645$$



Part (c)

If the observed sample mean is $\bar{x}$, then the observed test statistic is

$$Z_{\text{obs}} = \frac{\bar{x} - 0}{\sigma/\sqrt{n}}$$

Since this is a right-tailed test ($H_1: \mu > 0$), the p-value is the probability, assuming H_0 is true, of observing a value of the standard normal test statistic at least as large as Z_{obs} . Thus,

$$p\text{-value} = P(Z \geq Z_{\text{obs}}) = 1 - \Phi(Z_{\text{obs}}), \quad Z \sim N(0, 1)$$

A small p-value means that the observed sample mean is unusually large under $H_0: \mu = 0$. Therefore, the data provide evidence against H_0 and in favor of $H_1: \mu > 0$. It does not mean that the probability H_0 is true is small.

Problem 5

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$. Consider the following hypothesis test:

$$H_0 : p = 0.3 \text{ vs. } H_1 : p = 0.6$$

- a) (2 points) Write out the likelihood ratio:

$$\Lambda(x) = \frac{L(0.3)}{L(0.6)}$$

- b) (2 points) Using the Neyman–Pearson Lemma, explain why the most powerful test rejects for large values of $\sum_{i=1}^n X_i$.
- c) (3 points) Suppose the rejection rule is:

$$\sum_{i=1}^n X_i \geq c$$

Explain conceptually how power depends on the choice of c .

- d) (3 points) Suppose the rejection region is $\sum_{i=1}^n X_i \geq c_n$ where,

$$\frac{c_n}{n} \rightarrow k \quad \text{for some } 0.3 < k < 0.6$$

Is this sequence of tests consistent against the alternative $p = 0.6$? In other words, does the power converge to 1 as $n \rightarrow \infty$?

Page intentionally left blank

(a) Let $S = \sum_{i=1}^n X_i$ Since $X_1, \dots, X_n$ are iid Bernoulli(p), the likelihood is

$$L(p|x) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

Therefore,

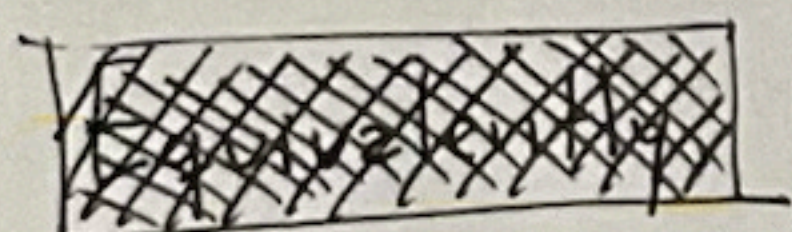
$$L(p|x) = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i} = p^S (1-p)^{n-S}$$

$$\text{Under } H_0: p=0.3, \quad L(0.3|x) = 0.3^S 0.7^{n-S}$$

$$\text{under } H_1: p=0.6, \quad L(0.6|x) = 0.6^S 0.4^{n-S}$$

Hence the likelihood ratio is

$$\Lambda(x) = \frac{L(0.3|x)}{L(0.6|x)} = \frac{0.3^S 0.7^{n-S}}{0.6^S 0.4^{n-S}} = \left(\frac{0.3}{0.6}\right)^S \left(\frac{0.7}{0.4}\right)^{n-S}$$



$$(b) \text{ From (a)} = \left(\frac{0.3}{0.6}\right)^S \left(\frac{0.7}{0.4}\right)^{n-S} \longrightarrow \Lambda(x) = \left(\frac{1}{2}\right)^S \left(\frac{7}{4}\right)^{n-S} = \left(\frac{7}{4}\right)^n \left(\frac{2}{7}\right)^S$$

Since $\frac{2}{7} < 1$, the term $\left(\frac{2}{7}\right)^S$ decreases as S increases. Therefore, $\Lambda(x)$ is a decreasing

function of $S = \sum_{i=1}^n X_i$

By the Neyman-Pearson Lemma, the most powerful level- α test rejects H_0 when

$$\Lambda(x) \leq K \quad \text{some } K. (\text{constant})$$

Since $\Lambda(x)$ decreases as S increases, this rejection is equivalent to rejecting for large values of S . Hence the most powerful test has rejection region of the form

$$\text{Reject } H_0 \text{ when } \sum_{i=1}^n X_i \geq c \quad \text{for some cutoff } c.$$

(c) Let $S = \sum_{i=1}^n X_i$, The rejection rule is Reject H_0 if $S \geq c$.

Since under the alternative $p = 0.6$, we have $S \sim \text{Binomial}(n, 0.6)$, the power of the test is $\pi(0.6) = P_{0.6}(S \geq c)$. Conceptually, the power depends on c as follows:

- If c is increased, the rejection region becomes smaller, so it becomes harder to reject H_0 , Therefore: Larger $c \rightarrow$ Smaller power.
- If c is decreased, the rejection becomes larger, so it becomes easier to reject H_0 . Therefore: smaller $c \rightarrow$ larger power.

However, lowering c also increases the probability of rejecting H_0 when H_0 is true, so there is a trade-off between power and Type I error.

Thus, conceptually, power is the probability under $p = 0.6$ that the number of successes is at least c , and this probability decreases as c increases.

(d) Let $S_n = \sum_{i=1}^n X_i$. The rejection rule is $S_n \geq c_n$.

Dividing by n , this is equivalent to $\bar{X}_n = \frac{S_n}{n} \geq \frac{c_n}{n}$

So the power under the alternative $p = 0.6$ is $P_{0.6}(S_n \geq c_n) = P_{0.6}\left(\bar{X}_n \geq \frac{c_n}{n}\right)$

Since $X_1, \dots, X_n$ are iid Bernoulli(0.6) under the alternative, the Law of Large Numbers gives $\bar{X}_n \xrightarrow{P} 0.6$. We are also given that $\frac{c_n}{n} \rightarrow k$ with $0.3 < k < 0.6$

Because $k < 0.6$, the rejection cutoff converges to a number strictly smaller than the true alternative value. Therefore, for large n , the event $\bar{X}_n \geq \frac{c_n}{n}$ occurs with probability tending to 1.

Hence $P_{0.6}(S_n \geq c_n) \rightarrow 1$. So the sequence of test is consistent against the alternative $p = 0.6$. So, Yes, the power converges to 1, so the test is consistent.